

Travel Demand Forecasting

The Four-Step Model: Trip Generation & Distribution

Lewis Lehe

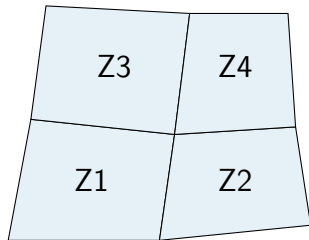
Fall 2025

Why forecast travel?

- Transportation infrastructure decisions are **long-lived** and **expensive**: highways, transit lines, bridges last decades.
- Planners need to answer questions like:
 - How much traffic will this new highway carry in 2050?
 - Will a new rail line attract enough riders?
 - How will land-use changes affect congestion?
- In the U.S., **Metropolitan Planning Organizations (MPOs)** are required to maintain regional travel demand models.
- The **four-step model** has been the workhorse approach since the 1950s.

Traffic Analysis Zones (TAZs)

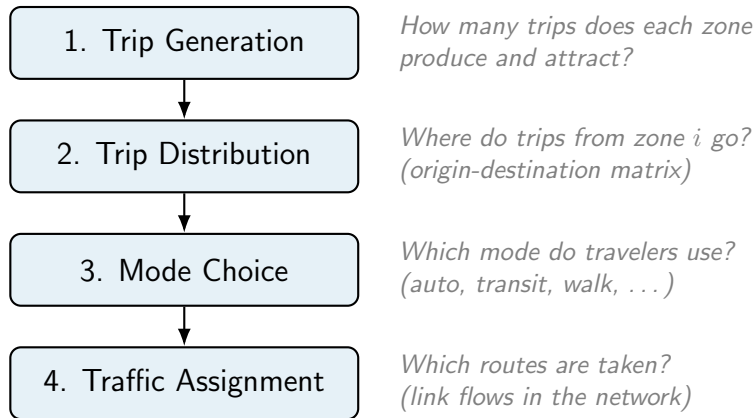
- The region is divided into **Traffic Analysis Zones (TAZs)**.
- Each zone aggregates information about:
 - Households (population, income, autos)
 - Employment (jobs by sector)
 - Land use (retail, schools, etc.)
- Trips between zones are what we forecast.
- A typical metro region has hundreds to thousands of TAZs.



a small region

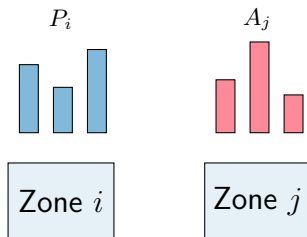
The Four Steps

The four-step model



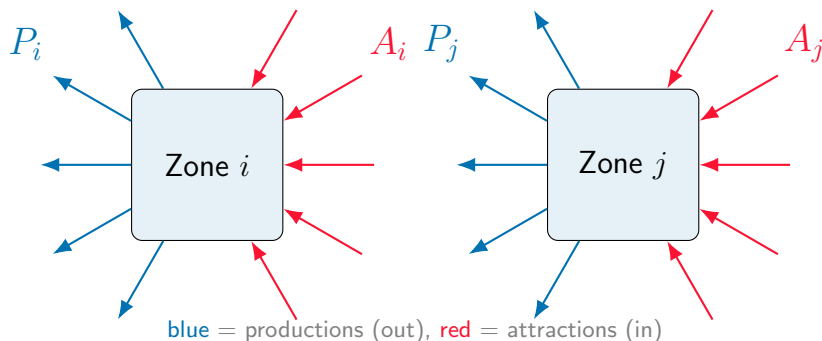
Stage 1: Trip Generation

- For each zone, predict:
 - **Productions** P_i : trips originating in zone i
 - **Attractions** A_j : trips ending in zone j
- Inputs: zone population, jobs, income, autos.
- Output: two columns of numbers, one per zone.



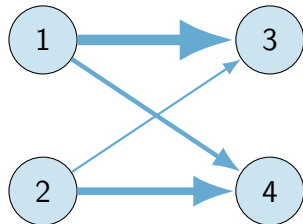
Stage 1: Trip Generation, visually

- Each zone *produces* P_i trips (out) and *attracts* A_j trips (in).



Stage 2: Trip Distribution

- Given P_i and A_j , decide *where each zone's trips go*.
- Output: an **origin-destination (O-D) matrix** T_{ij} .
- Row sums: $\sum_j T_{ij} = P_i$.
- Column sums: $\sum_i T_{ij} = A_j$.

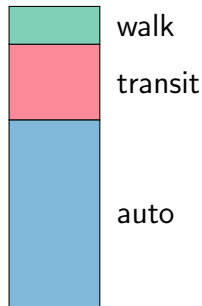


thicker = more trips

T_{ij}	3	4
1	8	2
2	1	5

Stage 3: Mode Choice (covered next time)

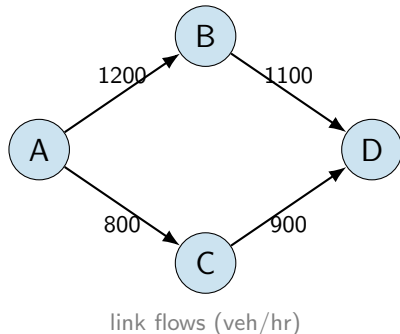
- For each O-D pair, split trips by **mode**: auto, transit, walk, bike, ...
- Usually modeled with **discrete-choice** (logit) models based on travel time, cost, and demographics.
- *We will cover utility theory and the logit model in the next lecture.*



mode shares for one O-D pair

Stage 4: Traffic Assignment (covered next time)

- Given the auto trips for each O-D pair, decide *which routes* they take through the network.
- Travel time on a link depends on its flow (congestion).
- Solved with **Wardrop's user equilibrium** principle.
- *Next lecture: BPR functions, Wardrop's principles, and Braess's paradox.*



Recap: inputs and outputs of each stage

- 1. Trip Generation

- In: zone demographics and land use
- Out: P_i , A_j (trip totals per zone)

- 2. Trip Distribution

- In: P_i , A_j , inter-zone travel impedance t_{ij}
- Out: T_{ij} (O-D matrix)

- 3. Mode Choice

- In: T_{ij} , level-of-service for each mode
- Out: T_{ij}^m (O-D matrix for each mode m)

- 4. Traffic Assignment

- In: T_{ij}^{auto} , network with link cost functions
- Out: link flows and travel times

- Today we focus on Steps 1 and 2.

Trip Generation

Productions and attractions

- A **trip** is a one-way movement from origin to destination.
- **Productions** are usually associated with the *home* end of a trip.
 - Households produce trips (commuting, shopping, school).
- **Attractions** are associated with the *non-home* end.
 - Workplaces, schools, retail, hospitals attract trips.
- Trips are stratified by **purpose**:
 - HBW (home-based work)
 - HBO (home-based other: shopping, school)
 - NHB (non-home-based)
- Different purposes have different generation rates and impedance sensitivity.

Where do the trip rates come from?

- Both methods need empirical data on *who travels how much*.
- In the U.S., the canonical source is the **National Household Travel Survey (NHTS)**, run by the Federal Highway Administration.
 - Conducted roughly every 5–8 years (1969, 1977, . . . , 2009, 2017, 2022).
 - Households fill out a 24-hour **travel diary**: every trip, mode, purpose, time, and distance.
 - Plus demographics: household size, income, vehicles, workers.
- Many MPOs also run their **own regional household travel surveys** for local calibration.
- Output: trip rates by category, trip-length distributions, mode shares — the inputs to all four steps.

Method 1: Regression

- Treat each zone as one observation. Fit a linear model:

$$P_i = b_0 + b_1X_{i1} + b_2X_{i2} + \cdots + b_kX_{ik} + \varepsilon_i$$

- X_{ik} are zone-level explanatory variables, e.g.:
 - households in zone i
 - population
 - autos owned
 - median income
- Coefficients b_k are estimated by **ordinary least squares** on a base-year survey.
- A separate regression is fit for each trip purpose.
- For forecasting: plug future X_{ik} into the equation to get future P_i .

Regression: strengths and limitations

- Strengths

- Compact: a few coefficients summarize the whole region.
- Easy to extrapolate to zones not in the original sample.
- Standard regression diagnostics apply.

- Limitations

- Aggregating households into zonal averages **loses information** about within-zone variation.
- Variables like population, households, and autos are highly **correlated** (multicollinearity).
- Linear form may not capture saturation effects (e.g., trip-making per household plateaus).

Method 2: Cross-Classification

- Stratify *households* by category, e.g. household size $\times$ autos owned.
- For each cell, compute an empirical **trip rate** (trips/household/day) from a household travel survey.
- Trip rates are assumed *transferable* across zones with similar populations.

	0 autos	1 auto	2+ utos
1 person	1.2	2.4	2.8
2–3 persons	2.6	4.5	6.1
4+ persons	3.8	6.0	7.9

trips/household/day (illustrative)

Cross-classification: applying the rates

- For each zone i , count the households in each category c : H_{ic} .
- Predicted productions:

$$P_i = \sum_c r_c \cdot H_{ic}$$

where r_c is the trip rate from the table.

- For forecasting: predict the future household distribution by category, then apply the same rates.
- **Key assumption**: trip rates per household-type stay roughly constant over time (much weaker than assuming linear coefficients are constant).

Regression vs. cross-classification

- **Regression** is parametric: assumes a functional form (linear), uses few parameters.
- **Cross-classification** is non-parametric: makes no functional-form assumption, but needs enough survey data for *each* cell.
- Cross-classification handles **nonlinearity** naturally (e.g., big jump from 0 to 1 auto).
- Regression **transfers** more easily to new regions; cross-classification needs cell-level data there too.
- In U.S. practice today, **cross-classification dominates trip generation** for households; regression is sometimes used for attractions (employment-based).

Trip Distribution

From totals to an O-D matrix

- Trip generation gave us *row totals* P_i and *column totals* A_j .
- Trip distribution fills in the cells T_{ij} .

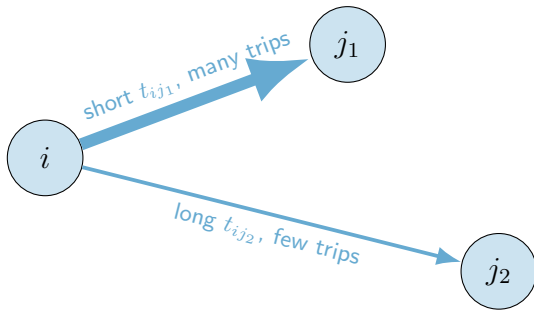
$$\sum_j T_{ij} = P_i, \quad \sum_i T_{ij} = A_j$$

- Many matrices satisfy these constraints — which one is right?
- Use a **behavioral model**: people prefer destinations that are *closer* (lower travel time / cost).

	$j = 1$	$j = 2$	$j = 3$	P_i
$i = 1$	T_{11}	T_{12}	T_{13}	1000
$i = 2$	T_{21}	T_{22}	T_{23}	800
$i = 3$	T_{31}	T_{32}	T_{33}	600
A_j	700	1100	600	

Intuition: trips fall with impedance

- Let t_{ij} be the **impedance** between zones i and j (travel time, distance, or generalized cost).
- Empirically: as t_{ij} grows, the share of i 's trips that go to j **shrinks**.
- Two destinations equally attractive in A_j but at different distances do not get equal flows — the closer one wins.



The gravity model

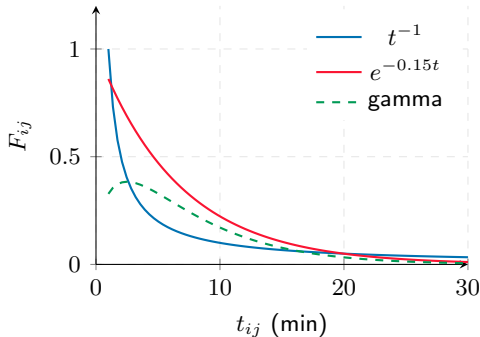
- The **gravity model** distributes P_i across destinations in proportion to “attractiveness divided by impedance”:

$$T_{ij} = P_i \cdot \frac{A_j F_{ij} K_{ij}}{\sum_k A_k F_{ik} K_{ik}}$$

- Components:
 - P_i : productions in i (from Step 1)
 - A_j : attractions in j (from Step 1)
 - F_{ij} : **friction factor**, a decreasing function of t_{ij}
 - K_{ij} : socioeconomic adjustment factor (often $K_{ij} = 1$)
- By construction, $\sum_j T_{ij} = P_i$ for every i — so **row sums are automatically satisfied**.

Friction factor: common forms

- **Power:** $F_{ij} = t_{ij}^{-n}$
- **Exponential:** $F_{ij} = e^{-\beta t_{ij}}$
- **Gamma:** $F_{ij} = a t_{ij}^b e^{-ct_{ij}}$
- Parameters are calibrated so the *predicted trip-length distribution* matches the survey.



Why “gravity”?

- Newton's law of gravitation: force $\propto m_1 m_2 / d^2$.
- Original (unnormalized) trip-distribution analogy:

$$T_{ij} \propto \frac{P_i A_j}{t_{ij}^2}$$

- Modern form replaces $1/t^2$ with a general friction factor F_{ij} , and **normalizes** so that row sums equal P_i .
- The shares interpretation makes it a proper conservation law for productions.

Doubly-constrained gravity model

- The basic gravity formula guarantees row sums but **not** column sums.
- In general, $\sum_i T_{ij} \neq A_j$.
- The **doubly-constrained** version introduces row and column balancing factors a_i, b_j :

$$T_{ij} = a_i b_j P_i A_j F_{ij}$$

with a_i, b_j chosen so that both P_i and A_j are matched.

- Solved by **iterative proportional fitting** (the Furness method): alternately scale rows to P_i , then columns to A_j , until convergence.
- *Details are covered in CEE 417 (Transportation Planning).*

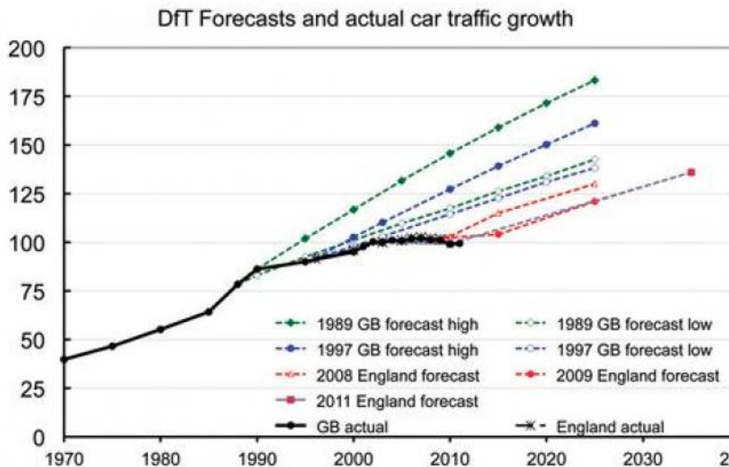
Recap

- The **four-step model** chains four sub-models: generation $\rightarrow$ distribution $\rightarrow$ mode $\rightarrow$ assignment.
- **Trip Generation** produces zonal P_i and A_j . Two main methods:
 - **Regression**: linear model on zone variables.
 - **Cross-classification**: empirical trip rates per household category.
- **Trip Distribution** builds the O-D matrix T_{ij} using the **gravity model**, with a friction factor F_{ij} that decreases in impedance.
- Singly-constrained gravity matches P_i ; doubly-constrained also matches A_j via balancing.

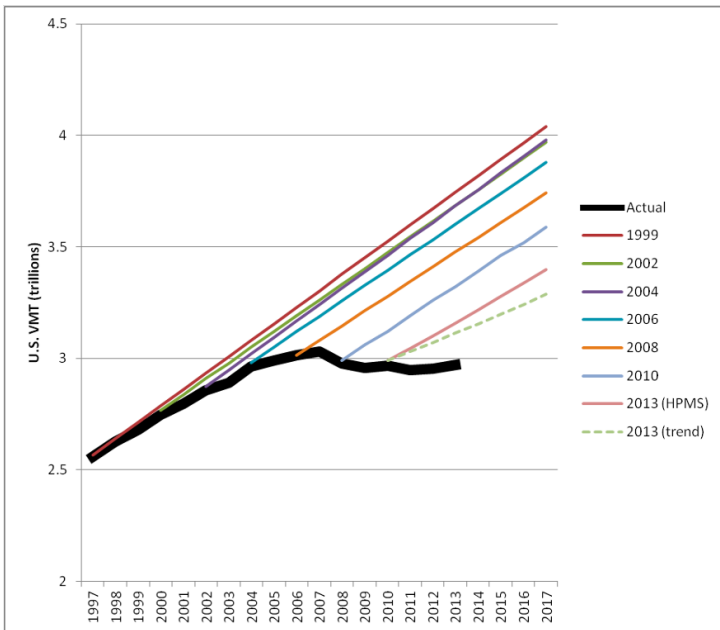
Next time

- **Mode Choice:** utility theory and the logit model — how do we split T_{ij} across modes?
- **Traffic Assignment:** Wardrop's user equilibrium, the BPR function, and Braess's paradox — how do auto trips spread across the network?
- Reading: review the four-step framework in your planning text; we will build on the same notation.

Be Skeptical



Be Skeptical



Be Skeptical

Should we trust WSDOT Traffic Projections?

Actual weekday traffic on SR-520 vs. WSDOT forecasts.

